Predicting NBA Players Future Value Over Replacement Player using Deep Learning

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A dissertation presented in part fulfillment of the requirements of the Degree of Master of Science at the University of Glasgow

13 December 2021

Abstract

National Basketball Association (NBA) is one of the most profitable, popular, and marketable basketball leagues in the whole world. Subsequently, it has a wealth of statistical performance data that can be studied to predict outcomes that can help elevate the status of the game. In this research study, we aim to improve upon the existing state of the art problem published in 2021 (Nguyen Hoang Nguyen, 2021) involving supervised Deep Learning approaches to predict the performance of NBA players whilst also suggesting a new metric according to which a player's performance should be judged. The newly suggested metric, value over replacement player is also compared against the existing performance metric, Win Shares. In our results, Deep Learning performs well on a data structure with an increased number of predictor variables when compared to the previous study and 2 layered Neural Network best improves the existing state of the art MSE value by 4.2890. It also indicates that the proposed judgement metric, "Value Over Replacement Player" can produce better results than the "Win Shares" as the least predicted MAE and MSE value in VORP prediction are 0.0354 and 0.0022 respectively.

Education Use Consent

I hereby give my permission for this project to be shown to other University of Glasgow students and to be distributed in an electronic form.

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Acknowledgements

I want to thank my supervisor, Dr José Cano Reyes for his continuous support and weekly scheduled meetings. His guidance and confidence were groundbreaking for my research study that not only helped me to complete my project but also elevated my interests in the research tasks.

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Chapter 1 Introduction

This chapter describes the introduction to the concept of the study, motivation that made us choose this project and the objectives we want to achieve from undertaking this research project along with the structure of the report.

National Basketball Association (NBA) is one of the major sporting events played in America and it is the world's most prestigious men's professional basketball league (Rathborn, 2020).The league comprises of 30 teams, 29 from different states in America and 1 from Canada, participating in an annual event that runs from October to April. It is also regarded as one of the most profitable leagues globally as the teams generate a revenue of about 209 million US dollars, ranging up to 474 million US dollars for the highest-earning teams (Gough, 2021). In addition, a team's revenues are also affected by the advertisement and endorsements of the different players playing for it.

1.1 Motivation

Each participating team roster contains 15 players who play in an 82-game long season. These players represent their state whilst playing for their team and form the core part of an organization. An organization's performance, popularity and reputation are largely dependent on how their players perform in an upcoming season. The performance of several players (normally known as rotation) decides whether their city would be the champion this year. This indicates if they will be able to make it to the playoffs or finish last in the regular-season rankings. Hence, predicting a player's performance efficiently is one of the key tasks that can be performed. This would help an organization make several decisions, such as informing them that a player is highly valuable to the team and hence should be offered a large sum of money in their next year's contract. Additionally, coaches can also be notified that a specific player's performance is decreasing, and supplementary drills can be introduced for him that will help elevating his game.

1.2 Aim and Objective

This project aims to improve a Deep Learning model's performance in predicting an NBA player's performance, measurable by statistics. We build upon the existing state of the art model of (Nguyen Hoang Nguyen, 2021) and present a comparative study about how a Deep Learning model's ability to perform a regression task could be enhanced so that a player's performance can be judged and predicted effectively. For this reason, we suggest a new metric according to which a player's performance should be evaluated.

1.2.1 Value Over Replacement Player (VORP)

We researched what the best performance measuring statistic should be that would be representative of a player's performance. Conclusively, using a player's value over a replacement player (VORP) is the best indicator of how a player

performs on both ends of the floor for an entire 82 game season. It transcends a player's value to the team over what a league's average player should be. According to statisticians, it was established that -2.0 is the replacement level in the NBA and any value above that metric would represent how valuable a player would be compared to that replacement-level player (Myers, 2020).

1.2.2 Advantages of Using VORP

a) Along with using the traditional box score statistics, calculation of VORP transcends BPM that uses concepts like Regression Basis to offer the best estimate of a player's contribution to a team and further considers the player's playing time. This is the reason it is favored over other metrics such as Box Plus-Minus (BPM) or Win Shares.

b) Converts the BPM rate into an assessment of each player's overall contribution to the team in comparison to a hypothetical "replacement player."

c) Tracks linearly with salary; hence, salary for the upcoming season can also be predicted if the difference in VORP is known.

d) Wins over Replacement can also be easily calculated if VORP is known and hence would help find how much a team would be affected if the player is traded, injured, retired, etc.

1.3 Report Structure

This report is presented in 5 chapters.

- The first chapter describes the introduction to the concept, motivation that made us choose this project and the objectives we want to achieve from undertaking this research project along with the structure of the report.
- The second chapter analyzes the previous works in this field and a critical overview of the state-of-the-art problem that we want to improve upon.
- The third chapter describes the data sources, the different methodologies that were used to conduct this study and how the design is implemented to produce the results.
- The evaluation chapter forms the fourth chapter and contains the results of the study for both validation and test datasets.
- Finally, the main conclusions of the study and suggestions for further work are presented in the final chapter.

Chapter 2 Survey

Almost every NBA team has a separate section that is dedicated to incorporating different techniques such as machine learning algorithms, data analysis, to design a better winning strategy, predicting and avoiding player injury along with generating an effective scouting report to create a better class of first year players. The application of Data mining algorithms including Machine Learning (ML) for the prediction of results of various sports has been carried out since the 1980s. Year by year descriptions along with some notable critiques of works are presented here. This chapter has been further categorized into two sections, where the former mainly discusses a historical evaluation of studies in this field while the latter mentions the analysis of current study in this area.

2.1 Background Concepts

In the 1980s, there have been works for calculating descriptive and statistical features. Earlier stages of usage of Data Analysis can be demonstrated through Advanced Scout (AS) software produced by (Bhandari, I., Colet, E., Parker, J. et al., 1997) which helped a coach assess different plays to increase the effectiveness of certain coaching decisions and formulate game strategies for subsequent games. The probability of opposing team player to either go for a drive, pull up for a jump shot or pass the ball were primarily assessed. Findings of this study included results like "When Price was Point-Guard, J. Williams missed 0% (0) of his jump field-goal-attempts and made 100% (4) of his jump field-goal-attempts. The total number of such field-goal attempts was 4. This is a different pattern than the norm which shows that: Cavaliers players missed 50.70% of their total field-goal-attempts. Cavaliers players scored 49.30% of their total field-goal-attempts." (Bhandari, I., Colet, E., Parker, J. et al., 1997). In the 1995-96 season, AS software was distributed to sixteen of the twenty-nine NBA teams. This software's main goal was to find intriguing patterns in basketball game data. Thus, it aided coaches in evaluating results to make better coaching judgments and develop a productive game strategy for forthcoming games. It was well received by the NBA because this data collection improved the quality of play. Nevertheless still, the study could not answer whether an individual performance could change the outcome of the game. In 2001, the study by (Melnick, 2001) aimed to prove a correlation between assists per game and the outcome of the game. However, this study just focused on how the unselfish play of players affects a game and there was no concrete evidence about how it was affecting a player's performance. In the year 2012, the study (Cao, 2012) used a different set of data for each team for both, offence and defense and aimed to determine if an NBA team will face a win or a loss. He applied several ML algorithms, including Simple Logistics Classifier, Artificial Neural Networks (ANN), Support Vector Machines (SVM) and Naïve Bayes (NB) and achieved the best prediction accuracy of 69% by Logistic Regression (LR). This accuracy was significant but still couldn't answer if a ML algorithm can predict how individual player would perform. Further, as the NBA started incorporating more player tracking cameras, the analysis techniques were deepened. Traditional methods of analysis that were concentrated on the end results of possession got replaced with a model that qualitatively evaluates every aspect of a play. Every decision that could be taken by a player ranging from the induvial passes to every

teammate, probability of a defender to either go right or left, how many times a player falls for a fake, what is the best attribute of a certain action in each possession were analyzed. Conventional terms to measure stats such as Field Goal Percentage (FG %) were replaced with Effective Field Goal Percentages (EFG%) that considered how effective a certain field goal is. As shown in the study by (Cervone, Dan and D'Amour, Alexander and Bornn, Luke and Goldsberry, Kirk, 2014), NBA analysts started modelling hypothetical average player with a certain decision-making ability and a skillset and used it as a model to improve the performance of teams and evaluate a player they might need in the upcoming draft prospects.

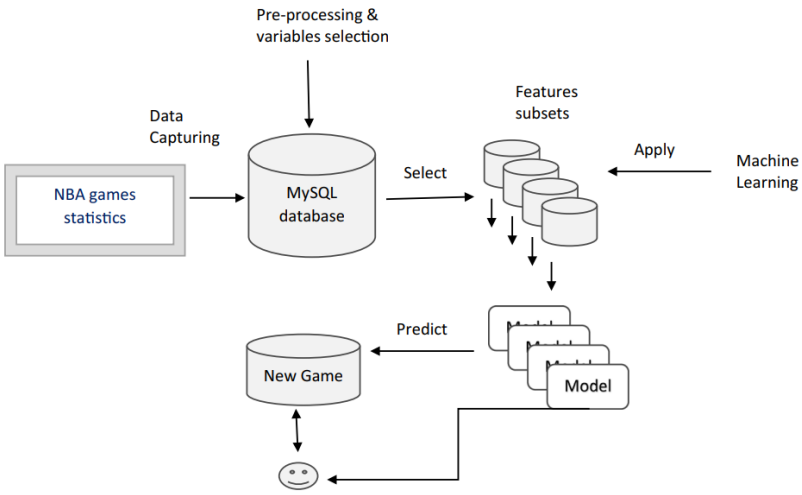


Figure 1: Proposed ML framework (Fadi Thabtah, 2019)

(Fadi Thabtah, 2019) incorporated Feature Analysis along with ML frameworks to predict the results of the games played at the NBA which can be seen in the figure 1. This was done by discovering the influential feature set that affected the outcome of the games. First, through the study of historical data (1980-2017) significant factors that affect the outcome of the games were found. Next, the prediction accuracy was maximized by utilizing the found set of features. The main Machine Learning models used were simple ANN, NB, and Logistic Model Tree (LMT). The Feature reduction techniques used were: Two filter methods (Multiple Regression and Correlation Feature Set) and One Rule induction algorithm (Ripper algorithm). In conclusion, the set of features that were most influential to NBA games was found. Also, they found that removing some features was increased the accuracy for all the models apart from ANN. However, applying Deep Learning (DL) to the win predictions was one of the improvements that could have been made to obtain more concrete results.

(Minčev, 2021) performed a case study in National Hockey League (NHL) player salary prediction by incorporating the SEMMA (Sample, Explore, Modify, Model and Access) process to yield a predictive data mining model. The models used were dummy regressor, Linear regression, k-nearest neighbor (KNN), Decision Tree (DT), Random Forest (RF), Neural Networks (NN). Through the low average and mean absolute error it optimally highlighted that if it is used effectively in NHL, "sporting organizations have the power to leverage data to develop a

competitive advantage" (Minčev, 2021). Furthermore, (Kaimakamis, 2021) applied several Machine Learning algorithms which measured every team's game performance and predicted the final position after the NBA playoffs. For every season they analyzed the regular season performance and predicted the NBA champion that season. The main aim was to train the models to make the most accurate prediction using team statistical categories from 18 seasons of NBA. Although the study could produce a considerable number of precise predictions, it was not free of inaccurate ones. However, several interesting findings were presented in the study as nearly every classifier predicted Los Angeles Lakers to be crowned as the NBA champion, which was confirmed as the Lakers went on to win the championship that year.

As Deep Learning was included more in Sports predictions, the study by (Huang, 2021) incorporated Deep Learning methods along with ML to predict the outcome of Major League Baseball matches. They collected the box score data of 30 teams from the 2019 season and used a one-dimensional convolutional neural network (1DCNN), ANN and SVM for the predictions. The prediction findings indicated that when all pitchers' data was taken into account, the models had greater forecast accuracy than when only the SP's data was considered. These three proposed models had a higher accuracy than any other related studies. However, as the data used was only from the year 2019, the amount of available data was relatively small, which limited Deep Learning's prediction accuracy. The study could have been made more impactful if different feature selection methods were incorporated in the Data Mining process.

(Sarika Jain, 2021) incorporated DL models to predict the result of soccer games. Subsequently, there was an evaluation of the advantages and disadvantages of several techniques, followed by an effort to construct a prediction system powered by Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTMs). This study shows a visible advantage over the originally used ANN and traditional ML algorithms, by achieving a high evaluation accuracy of 81%. Nevertheless, it can be improved by using a better set of attributes.

2.2 State of the Art Analysis

Instead of focusing on game win predictions or champion predictions, (Nguyen Hoang Nguyen, 2021) primarily focused on predicting an NBA player's future performance and popularity. He applied several ML models including, LR, Gradient Boosting Algorithm (GBR), SVM, Polynomial SVM and NN, along with Deep Learning models, i.e., 2 Layered Network (2NN) and 3 Layered Neural Network (3NN) and 3 Layered Neural Network with increased nodes (3NN Increased). Impressive results were achieved for the ML models for both performance and popularity prediction, but DL performed poorly for predicting a player's performance, was due to certain issues.

Limitations in State of the Art

- The data presented was relatively small scale and structured with few predictor variables. This inhibited a DL model's capability to learn efficiently on the presented data which was noticeable through the high values of MSE while predicting a player's performance.

- Attempt to predict popularity along with the performance of NBA players was non-feasible for a regression-based model as predicting if a player will get selected in an all-star game and predicting his performance are two different tasks that don't correlate with each other, also were presented as two different tasks in the study.
- Features concerning "all-star" introduced a huge class imbalance for the regression tasks and led to many undesirable ramifications such as exorbitant MSE values for the test set even for the best performing DL models.
- The Removal of players who were traded in mid-season was also hurting a DL model as a player's performance is also dependent on the team for which they are playing. Just like in any team sport, team dynamics play a vital role in affecting the psychical and mental characteristics of a player which has a direct impact on a player's performance. As seen in the study by (Jordi Duch, 2010), there exists a strong correlation between the quality of team play and a player's performance. Hence, the players who got traded in mid-season shouldn't be removed from the dataset and should form separate instances for the model to learn and predict.

Chapter 3 Methodology

This section describes the sources, preprocessing and summary of the dataset used, along with the design and implementations of the model. The datasets were loaded and preprocessed with the help of Pandas and NumPy libraries in Python. All the models were designed with the help of Keras and tenserflow.

3.1 Data Sources

The unprocessed Dataset was obtained from Kaggle that contained 2 types of statistics of NBA players from 1950 to 2017. These two types of statistics (Lahman, 2021) are described in the following sections with the help of two tables.

3.1.1 Basic Box Score Attributes

In the earlier times, NBA primarily used the raw version of statistics such as the number of points scored by a player, the number of steals a player recorded, etc. This type of box score data only highlighted the basic attributes of a player instead of doing a comprehensive analysis of their play. The newer versions of basic box score attributes include the statistics such as eFG% which were able to transcend the player performance to a higher level than traditionally presented basic box score attributes. Table 1 presents the in-depth summary of the used basic box score attributes, whilst also presenting an insight of how they are calculated and what they represent.

Table 1: Basic Box Score Statistics

Statistics	**Full-Form**	**Description**
Year	Year	The year in which a particular NBA season ends. (2015-16 NBA season is treated as Year 2016)
Player	Player Name	Name of the Player
Position	Player Position	The basketball position at which the player plays
Age	Player Age	Age of the player as of February 1 of that year
Tm	Team	Team of the player
G	Games	Number of games the player played in a season

GS	Games Started	Number of games started by the player
FG	Field Goal	Number of field goals made by a player in a season (includes both 2 point and 3-point Field goals)
FGA	Field Goal Attempted	The number of field goals attempted by a player in a year
FG%	Field Goal Percentage	Field Goal Percentage of a Player. It is given by is FG / FGA.
3P	3 Pointers	The number of 3 pointers made by a player in a season
3PA	3 Pointers Attempted	3PT field goals attempted by a player in a season
3P%	3 Pointer Percentage	3PT Field Goal Percentage of a Player. It is given by is 3P /3PA
2P	2 Pointers	The number of 2 pointers made by a player in a season
2PA	2 Pointers Attempted	2PT field goals attempted by a player in a season
2P%	2 Pointer Percentage	2PT Field Goal Percentage of a Player. It is given by is 2P /2PA
eFG%	Effective Field Goal Percentage	Effective Field Goal Percentage of a player in a season. The formula is "(FG + 0.5 * 3P) / FGA". A 3-point field goal is worth one point more than a 2-point field goal, therefore this statistic adjusts for that.
FT	Free Throw	The number of Free Throws made by a player in a season

FTA	Free Throw Attempted	Free Throws attempted by a player in a season
FT%	Free Throw Percentage	Free Throw Percentage of a Player. It is given by is FT /FTA
ORB	Offensive Rebounds	Number of Offensive Rebounds taken by a player in a season
DRB	Defensive Rebounds	The number of Defensive Rebounds taken by a player in a season.
AST	Assists	Number of Assists by a player in a season
STL	Steals	Number of Steals by a player in a season
BLK	Blocks	Number of Blocks recorded by a player in a season
TOV	Turnovers	Number of Turnovers by a player in a season
PF	Personal Fouls	Number of Personal Fouls committed by a player in a season
PTS	Points	Number of Points scored by a player in a season

3.1.2 Advanced Stats

Advanced stats are a method of objectively analyzing basketball. It's a more in-depth approach to look at a simple box score and assess a player's or team's talent and output when on the court. (NBA, 2021) This detailed breakdown enables the data to capture minute factors of the game, such as how much impact a player is having on the team offensively and defensively. This helps the analysts and coaches to modify certain decisions and develop new strategies that will benefit a team or a player. The summary of different advanced statistics used in the study are described in the table.

Table 2: Advanced Statistics

Stat	**Full-Form**	**Description**
ORB %	Offensive Rebound Percentage	Estimate of a player's proportion of offensive rebounds while on the floor. The formula is "(ORB * (Tm MP / 5)) / (MP * (Tm ORB + Opp DRB))"
DRB %	Defensive Rebound Percentage	Estimate of a player's proportion of defensive rebounds while on the court. The formula is "(DRB * (Tm MP / 5)) / (MP * (Tm DRB + Opp ORB))"
TRB %	Total Rebound Percentage	Estimate of a player's proportion of total rebounds while on the court. The formula is "(TRB * (Tm MP / 5)) / (MP * (Tm TRB + Opp TRB))"
AST%	Assist Percentage	Estimate of team points that came through a player's assists while he was on the floor. The formula is "100 * AST / (((MP / (Tm MP / 5)) * Tm FG) - FG)"
STL%	Steal Percentage	Prediction of the opponents' possessions that result in a steal influenced by the player's while he was on the court. The formula is "(100 * (STL * (Tm MP / 5)) / (MP * Opp Poss))"
TOV%	Turnover Percentage	Estimate of turnovers made by a player per 100 plays. The formula is "(100 * (STL * (Tm MP / 5)) / (MP * Opp Poss))"

USG%	Usage Percentage	Estimate of percentage of team plays concerning the player. It is calculated by "(100 * ((FGA + 0.44 * FTA + TOV) * (Tm MP / 5)) / (MP * (Tm FGA + 0.44 * Tm FTA + Tm TOV))"
TS %	True Shooting Percentage	A measure of the player's shooting efficiency. It considers FG, 3P, 2P and FT. The formula is "(100 * ((FGA + 0.44 * FTA + TOV) * (Tm MP / 5)) / (MP * (Tm FGA + 0.44 * Tm FTA + Tm TOV))"
3PAr	3-point Attempt Rate	Percentage of field goals that came as 3P attempts by the player in a season.
FTr	Free throw Rate	Percentage of field goals that came as FT attempts by the player in a season.
PER	Player Efficiency Rating	Sums up a player's positive accomplishments subtracts the negative ones and returns a per-minute rating of a player's performance.
OWS	Offensive Win Share	An estimate of how many team victories that player's offensive performance contributed to.
DWS	Defensive Win Share	An estimate of how many team victories that player's defensive performance contributed to.

WS	Win Share	An estimate of how many team victories that player's overall performance contributed to.
WS/48	Win Share per 48 Mins	An estimate of how many team victories that player's overall performance contributed to per 48 minutes.
OBPM	Offensive Box Plus Minus	An estimate of the points per 100 possessions that a player provided to an average team by his offensive play over that of a league-average player.
DBPM	Defensive Box Plus Minus	An estimate of the points per 100 possessions that a player provided to an average team by his defensive play over that of a league-average player.
BPM	Box Plus Minus	An estimate of the points per 100 possessions that a player provided to an average team by his overall play over that of a league-average player.
VORP	Value Over Replacement Player	A prorated estimate of the points per 100 team possessions that a player provided over a replacement-level player, converted to an average team.

3.2 Dataset Preprocessing

As the data collected is raw and unfiltered it contains a high amount of noise, it must go through several data preprocessing tasks other than standard data cleaning steps to obtain the final and clean dataset. All the NaN values must be removed or managed. The various data processing steps are described below:

1) Removal of Years before 1980: - Almost all the stats were inconsistent before 1980, which to many NaN values for most features. Hence the years before 1980 are not considered and all the instances before that were removed.

2) Removal of "Games Started" Feature: - Due to a high number of instances with no consistent data on the Games Started by players, this feature was removed from the study.

3) Minimal Games: - It is not appropriate to predict performance based on a low number of games, i.e., players who can only play a few games (i.e., less than 5) in a season either because they were injured, traded, or benched for the entire season. Additionally, they also have incomplete stats in many categories and behave as noise rather than serving as optimal data for the model to learn on. Hence the players with less than 5 games were removed from the model training.

4) Zero 3 Point Attempts: - If a player has not attempted a single 3 Pointer, then his 3P% for that season should be 0 and hence NaN values for those instances must be replaced with 0.

5) Zero 2 Point Attempts: - If a player has not attempted a single 2 Pointer, then his 2P% for that season should be 0 and hence NaN values for those instances must be replaced with 0.

6) Zero Free Throw Attempts: - If a player has not attempted a single Free Throw, then his FT% for that season should be 0 and hence NaN values for those instances must be replaced with 0.

7) Player Name Correction: - Some players in the dataset have '*' attached to their name, which doesn't really serve a purpose and hence is removed from the Player Name feature. This task was performed with the help of code used in the previous study (Nguyen Hoang Nguyen, 2021).

8) Label Encoding for Player Position: - In a standard basketball game, the positions are labelled as 1 for PG, 2 for SG, 3 for SF, 4 for PF, 5 for C. Hence, for our categorical feature (POS), we encoded the positions according to standardized basketball game positions. Additionally, for players whose positions were split as they played more than one position, such as PG-SG, SG-SF, SF-PF, PF-C, it was assumed that they spent equal minutes for both positions and according to that they were imputed with an aggregate value of the positions.

9) One Hot Encoding for Categorical features: - The Deep Learning algorithms can't operate on labelled data directly and requires all input and output variables to be numeric. This requires encoding to be performed on our categorical features (Team and Player). As our data doesn't contain any ordinal relationship between our variables One Hot Encoding was chosen as the most logical method of encoding our categorical features.

3.3 Data Summary

In combination, these stats form 49 features, that encapsulate a player's performance and increase the number of features for a deep learning model to learn on. The two target variables used in the study, WS (Win Shares) and VORP (Value Over Replacement Player), are part of the same feature set, and the Deep

Learning models are evaluated according to some of the popular Machine Learning Evaluation Metrics:

1) Mean Absolute Error (MSE): - It calculates the magnitude of the difference between the prediction of observation and the true value of that observation.

2) Mean Squared Error (MAE): - The difference between the predicted values and true values are squared and averaged out across the whole dataset.

Correlation Analysis

The features used in the study are very extensive and as part of the standardized Machine Learning Data Mining Process, a heatmap was plotted to analyze how different features correlate to each other. The findings of the heatmap are described in the evaluation section.

3.4 Design

The models used in this study are 2 Layered Neural Network, 3 Layered Neural Network and 3 Layered Neural Network with an increased number of nodes. These deep learning models imitate the working of a human brain where the different layers can learn efficient representations of the seen data. The hyperparameters such as kernel initializer and activation function are kept identical to the last research so that an effective comparison can be made.

3.4.1 2 Layered Neural Network

A 2 layered Neural Network consists of 2 layers apart from an input layer and are connected sequentially. The kernel initializer was taken as normal whilst using ReLu as activation function for the initial layers. The optimizer used is Adam. Additionally, each layer has 256 neurons. The construction of a 2 Layered Neural Network with the help of Keras Sequential model can be seen in figure 2.

```
#2 layer model

sequential_two_vorp = Sequential()

# Input Layer :
sequential_two_vorp.add(Dense(128, kernel_initializer='normal',input_dim = X_train_original.shape[1], activation='relu'))

# Hidden Layer :
sequential_two_vorp.add(Dense(256, kernel_initializer='normal',activation='relu'))

# Output Layer :
sequential_two_vorp.add(Dense(1, kernel_initializer='normal',activation='linear'))

# Compile the network :
sequential_two_vorp.compile(loss='mean_squared_error', optimizer='adam', metrics=['mean_squared_error'])
sequential_two_vorp.summary()
```

```
Model: "sequential_1"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense_3 (Dense)             (None, 128)               354304

 dense_4 (Dense)             (None, 256)               33024

 dense_5 (Dense)             (None, 1)                 257

=================================================================
Total params: 387,585
Trainable params: 387,585
Non-trainable params: 0
_________________________________________________________________
```

Figure 2: 2 layered NN

3.4.2 3 Layered Neural Network

An additional layer is added in the middle for the construction of a 3 layered Neural network. Hence our model comprises 4 layers, an input layer, two layers in the middle and an output layer, which can be seen in figure 3. All the layers have the same number of neurons as 256, and the same activation functions and optimizers are used.

```
#3 layer model

sequential_three_ws = Sequential()

# The Input Layer :
sequential_three_ws.add(Dense(128, kernel_initializer='normal',input_dim = X_train_original.shape[1], activation='relu'))

# The Hidden Layers :
sequential_three_ws.add(Dense(256, kernel_initializer='normal',activation='relu'))
sequential_three_ws.add(Dense(256, kernel_initializer='normal',activation='relu'))

# The Output Layer :
sequential_three_ws.add(Dense(1, kernel_initializer='normal',activation='linear'))

# Compile the network :
sequential_three_ws.compile(loss='mean_squared_error', optimizer='adam', metrics=['mean_squared_error'])
sequential_three_ws.summary()
```

```
Model: "sequential_2"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense_6 (Dense)             (None, 128)               354304

 dense_7 (Dense)             (None, 256)               33024

 dense_8 (Dense)             (None, 256)               65792

 dense_9 (Dense)             (None, 1)                 257

=================================================================
Total params: 453,377
Trainable params: 453,377
Non-trainable params: 0
_________________________________________________________________
```

Figure 3: 3 layered NN

3.4.3 3 Layered Neural Network with Additional Nodes

When we increase the number of nodes in a 3 Layered neural network, our third model is obtained. This network essentially has the same number of layers as the previous model, but the nodes capturing the different aspects are increased from 256 to 512. This increment only takes place in the middle layers, the input and output layers are kept the same, as seen in figure 4.

```
#3 Layer model

sequential_three_increased_vorp  = Sequential()

# The Input Layer :
sequential_three_increased_vorp.add(Dense(128, kernel_initializer='normal',input_dim = X_train_original.shape[1], activation='rel

# The Hidden Layers :
sequential_three_increased_vorp.add(Dense(512, kernel_initializer='normal',activation='relu'))
sequential_three_increased_vorp.add(Dense(512, kernel_initializer='normal',activation='relu'))

# The Output Layer :
sequential_three_increased_vorp.add(Dense(1, kernel_initializer='normal',activation='linear'))

# Compile the network :
sequential_three_increased_vorp.compile(loss='mean_squared_error', optimizer='adam', metrics=['mean_squared_error'])
sequential_three_increased_vorp.summary()
```

```
Model: "sequential_5"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 dense_18 (Dense)            (None, 128)               354304

 dense_19 (Dense)            (None, 512)               66048

 dense_20 (Dense)            (None, 512)               262656

 dense_21 (Dense)            (None, 1)                 513

=================================================================
Total params: 683,521
Trainable params: 683,521
Non-trainable params: 0
```

Figure 4: 3 layered NN with Additional Nodes

We will fit the above-described models from the previous study by (Nguyen Hoang Nguyen, 2021) on both the predictor variables [WS and VORP] using WS results as the baseline for our model evaluations whilst also comparing the results from (Nguyen Hoang Nguyen, 2021).

3.5 Implementation

The designed models are implemented according to below described methods. This help in ensuring that a model is optimally trained and validated so that efficient evaluations can be made. Additionally, the random seed for every model is set to 1234 to ensure consistent results can be obtained.

3.5.1 Train Test Validation Split

To prevent the models from overfitting or underfitting, the whole data was split into two parts, train-validation data and testing data in the ratio of 70:30. Train-validation data was used to train the models with K-fold Cross-validation. The test dataset is used to obtain the final evaluation results for our models.

3.5.2 K-fold Cross-Validation

Cross-validation is a resampling method for preventing overfitting and measuring the quality of a machine learning model on unseen data by dividing the data into K parts. In each split, the model is trained on K-1 folds and validated on the other parts. The value of K must be such that each sample is large enough to be statistically representative of the broader dataset. As described in several ML and DL data mining algorithms, a standard value of 10 was chosen for K.

3.5.3 Repeated Evaluation Experiments

Models based on deep learning are inherently stochastic. This means that these models may be analyzed statistically but may not be predicted precisely. It happens because they contain random initial weights and random shuffling of data during each epoch. This generates different results each time even though when fit on the same data. Hence, to accurately estimate a model's skill and to control the model stability, we need to repeat the model evaluation several times and take the grand mean of the results. Therefore, we will repeat the model evaluation on Cross-validation 10 times and take the grand mean of our evaluation metrics to obtain the validation results.

```
mae_list_two_ws,mse_list_two_ws = [],[]

#Repeated evaluation experiments to control the uncertainity of the model
for i in range(10):
  mae_list,mse_list = [],[]

  #Performing Cross validation with 10 splits
  cv = KFold(n_splits = 10)
  for train_index, test_index in cv.split(X_train_original):
    X_train, X_test = X_train_original.iloc[train_index], X_train_original.iloc[test_index]
    y_train, y_test = y_train_original.iloc[train_index], y_train_original.iloc[test_index]


    #Model fitting with taken parameters
    sequential_two_ws.fit(X_train, y_train, epochs=10, batch_size=512, validation_split = 0, shuffle= True)
    curr_y = sequential_two_ws.predict(X_test)

    #Calculating the true and predicted values as a list
    y_true = y_test['WS'].values
    y_pred = curr_y.tolist()

    #Adding the CV Error for every split
    mae_list.append(mean_absolute_error(y_true,y_pred))
    mse_list.append(mean_squared_error(y_true,y_pred))

  #Adding the means to a list for calculating the grand mean
  mae_list_two_ws.append(np.mean(mae_list))
  mse_list_two_ws.append(np.mean(mse_list))
```

Figure 5: Model Implementation

As show in the figure 5, For 10 splits and n number of epochs, along with repeating the experiment for 10 times, in one run the model trains for 10*10*n times. For this reason, the number of epochs is also limited to 10. This ensures that the model is run enough times such that it trains and validates optimally according to the data whilst also not becoming too slow. Additionally, the batch size is kept as the same as the last study of 512, as this ensures that the model encapsulates every aspect of the dataset and contains the outliers (very high performing players).

Chapter 4 Evaluation

This chapter describes the different results obtained in the last run of performing the study. The different methodologies and the environment used to perform the evaluations are covered in Experimental Setup. Correlation Analysis of the different features are covered in the second section. Furthermore, the graphical results of validation set are presented in the third section. Finally, the test evaluations form the last section of this chapter. For the enhanced readability, all the error values are reported upto 4 decimal places.

4.1 Experimental Setup

The experiments were run on a local machine where model training validation and testing all took place to produce the results. The sequential models were trained with the help of Keras, and the different libraries used to perform the graphs and the heatmap are the visualization libraries of python, namely seaborn and matplotlib.

4.2 Correlation Analysis

From the heatmap presented in Figure 6, it can be observed that both of our predictor variables, namely WS and VORP have a strong correlation to the basic box score attributes and very little correlation to other advanced stats. This is explainable as calculation of both requires the arithmetic manipulation of a traditional scoresheet whilst not requiring the involvement of other advanced stats.

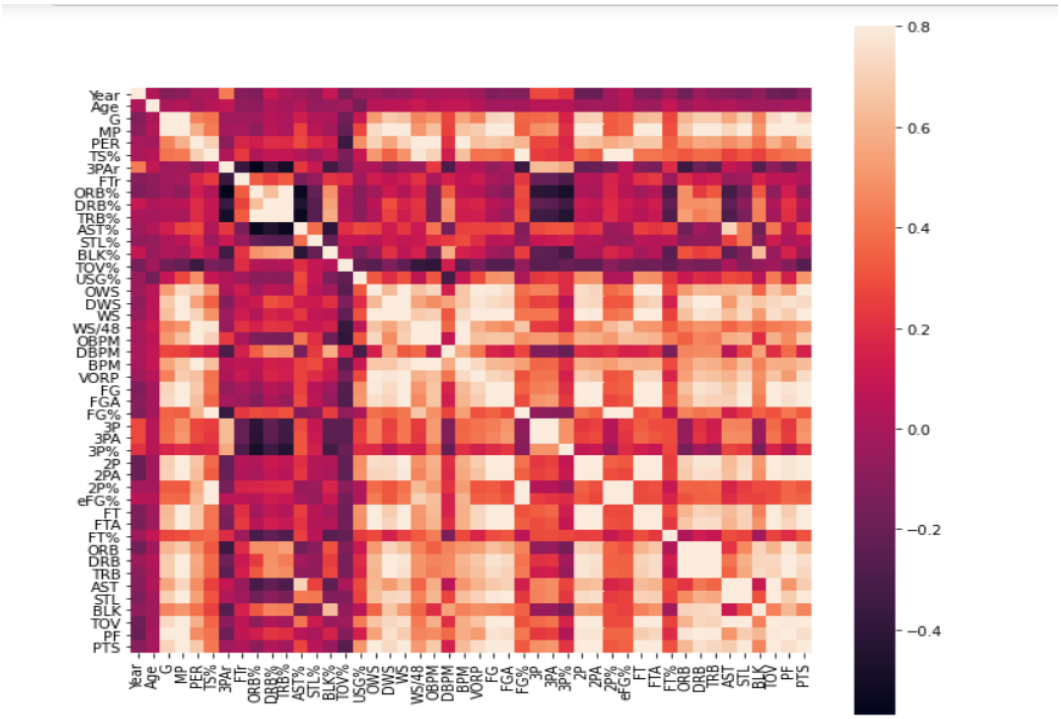


Figure 6: Heatmap

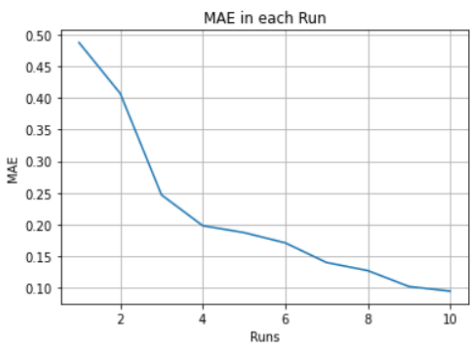
4.3 Validation Set Results

A graph was plotted to analyze how the validation error fluctuates in each run along with reporting the grand mean validation error. This was done for every model, on both the evaluation metrics. Additionally results from VORP were compared with the baseline results of WS.

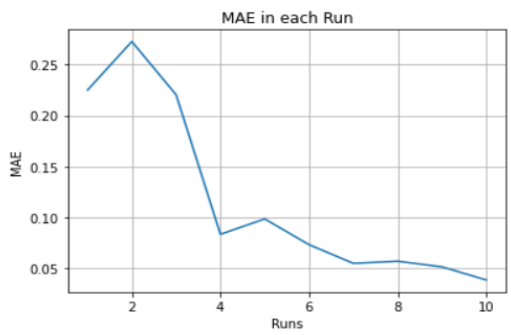
4.3.1 2 Layered Neural Network

A 2 Layer Neural Network is trained on the train-validation data obtained after splitting the dependent and independent variables and evaluated with the help of 10-fold Cross validation. The graphs below in Figures 7 and 8 describe the MAE and MSE in each run along with the validation error which were obtained by taking the mean of the validation results after repeating the experiment 10 times.

a) MAE: -



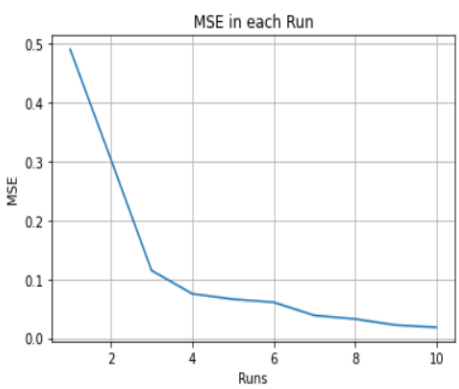
Validation MAE 2NN(WS): 0.216



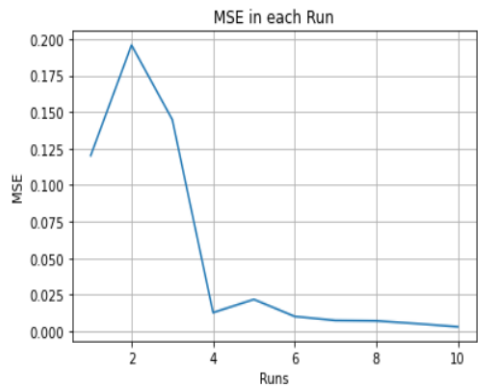
Validation MAE 2NN(VORP): 0.1175

Figure 7: Validation results for MAE of 2NN on WS and VORP

b) MSE: -



Validation MSE 2NN(WS): 0.1223



Validation MSE 2NN(VORP): 0.0528

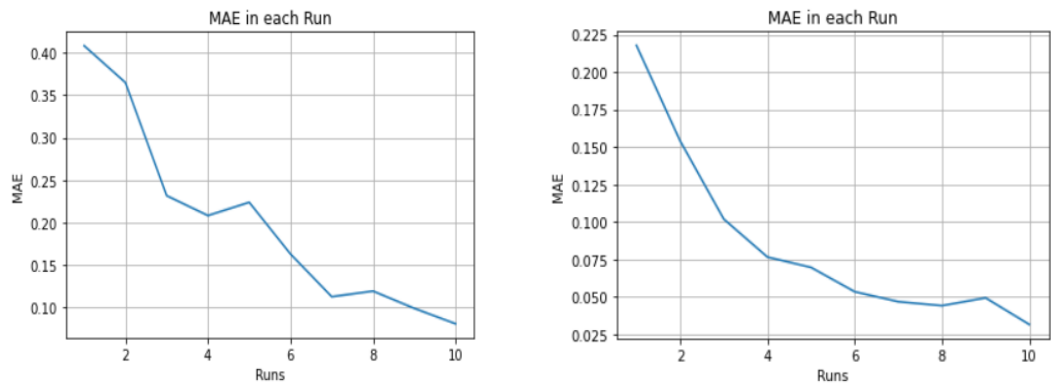
Figure 8: Validation results for MSE of 2NN on WS and VORP

We can observe from the graphs (Figure 7 and 8) that the model varies in every run for both the metrics, but generally follows a declining trend in error value with each run. Both MAE and MSE values on both the predictor variables are relatively low. However, the fluctuations when predicting VORP were sharper as compared to predicting WS, this was mainly because the error values were so low when predicting VORP that even a small change presented a sharp turn in the graph. In addition to that, our model also improves whilst predicting a player's value over replacement player (VORP) as compared to predicting his Win Shares (WS).

4.3.2 3 Layered Neural Network

The second model is also trained on the train-validation data obtained after splitting the dependent and independent variables and evaluated with the help of 10-fold Cross validation. The graphs below in Figures 9 and 10 describe the MAE and MSE in each run along with the validation error which were obtained by taking the mean of the validation results after repeating the experiment 10 times.

a) MAE: -

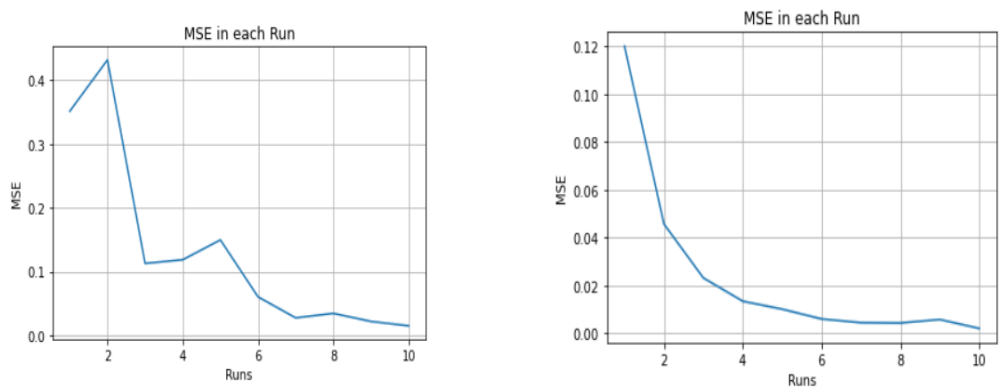


Validation MAE 3NN(WS): 0.2012

Validation MAE 3NN(VORP): 0.0846

Figure 9: Validation results for MAE of 3NN on WS and VORP

b) MSE: -



Validation MSE 3NN(WS): 0.1323

Validation MSE 3NN(VORP): 0.0234

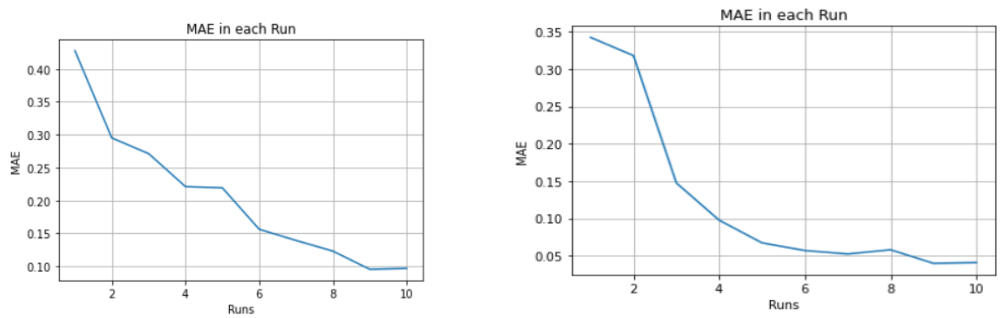
Figure 10: Validation results for MSE of 3NN on WS and VORP

Analyzing the graphs (Figure 9 and 10), we can see that the model is varying in every run for both the metrics, but like our first model follows a declining trend in error value with each run. Both MAE and MSE values on both the predictor variables are also relatively lower than the first model. This can be due to the addition of an extra layer may be improving a model capability to learn efficiently. However, we cannot say that surely without observing the test results. Moreover, just like the last model, the fluctuations when predicting VORP were sharper as compared to predicting WS, this was mainly because the values error values were much lower for VORP that even a small change presented a sharper turn in the graph. The In addition to that, our model also improves whilst predicting a player's value over replacement player as compared to predicting his Win Shares.

4.3.3 3 Layered Neural Network with Additional Nodes

Similar training and validation process occurred for the third model where it was trained on the train-validation data obtained after splitting the dependent and independent variables for both WS and VORP and evaluated with the help of 10-fold Cross validation. The graphs below in Figures 11 and 12 describe the MAE and MSE in each run along with the validation error which were obtained by taking the mean of the validation results after repeating the experiment 10 times.

a) MAE: -

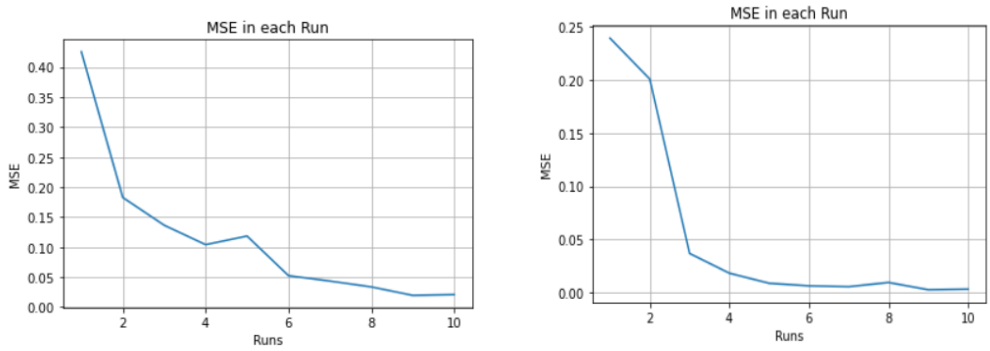


Validation MAE 3NN Increased(WS): 0.2045

Validation MAE 3NN Increased(VORP): 0.122

Figure 11: Validation results for MAE of 3NN Add. on WS and VORP

b) MSE: -



Validation MSE 3NN Increased(WS): 0.1136

Validation MSE 3NN Increased(VORP): 0.0532

Figure 12: Validation results for MSE of 3NN Add. on WS and VORP

From the above evaluations of the validation error (Figure 11 and 12), we can accurately estimate that all our models produce low values of both the errors. However, the introduction of additional nodes doesn't really improve the validation error, but again to obtain more concrete results, we will have to analyze the test observation of the models. Additionally, we can say that the model prediction becomes more accurate for every model for our chosen statistic relative to the baseline.

4.4 Test Results

Each model was used to predict the testset and predictions are compared with the actual values. This was done for both the predictor variables. Thereafter, the results are analyzed with the help of graphs and a pictorial and textual summarization of the error produced are presented here. Additionally, for the MSE error, we use the results from (Nguyen Hoang Nguyen, 2021) as a baseline to compare the results for WS. Moreover, comparisons are also made between WS and VORP to check if predicting VORP is increasing the accuracy of the model.

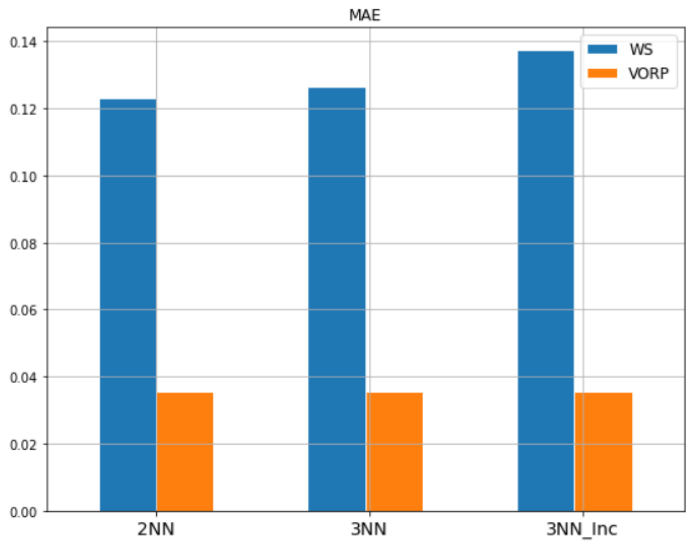
a) MAE: -

In test evaluation according to MAE (figure 13), the following significant results were obtained:

a) For the prediction of WS, the best performing model is 2NN with a minimum MAE of 0.1231. 3NN produces similar results and 3NN with Increased nodes produces an MAE value of 0.0142 higher than the best value.

b) For the prediction of VORP, both 3NN and 3NN_increased produce nearly identical results but 3NN gives the lowest value of 0.0354. These changes are hardly significant as the difference for 2NN is 0.0001 and for 3NN increased is 0.003. Hence, we can say that all our models are performing well on VORP.

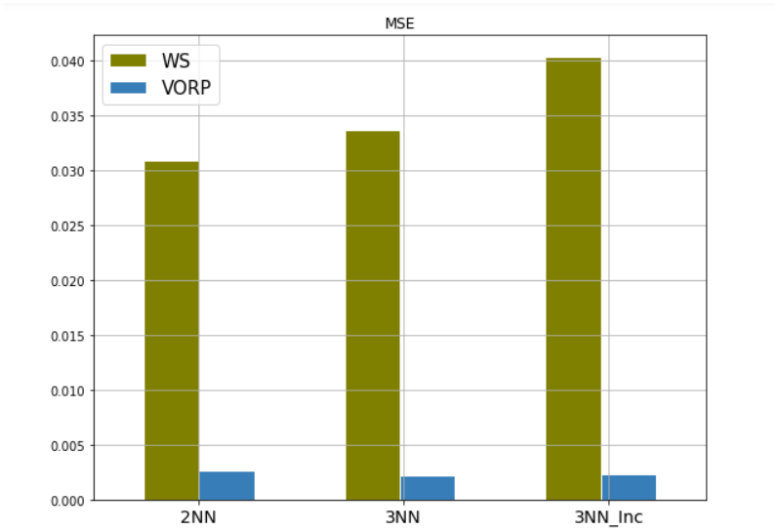
c) Every model improved its prediction accuracy when predicting a player's VORP when compared to predicting his WS. The average difference between the predictions is approximately 0.0876 for 2NN, 0.0910 for 3NN and 0.1016 for 3NN with increased nodes.




```
For 2NN(WS) MAE is 0.1231
For 2NN(VORP) MAE is 0.0355
For 3NN(WS) MAE is 0.1264
For 3NN(VORP) MAE is 0.0354
For 3NN Increased (WS) MAE is 0.1373
For 3NN Increased (VORP) MAE is 0.0357
```

Figure 13: Test MAE Results

b) MSE: -




```
For 2NN(WS) MSE is 0.031
For 2NN(VORP) MSE is 0.0027
For 3NN(WS) MSE is 0.0337
For 3NN(VORP) MSE is 0.0022
For 3NN Increased (WS) MSE is 0.0403
For 3NN Increased (VORP) MSE is 0.0023
```

Figure 14: Test MSE Results

For the test evaluation of MSE (figure 14), we obtained the following significant results:

a) For the prediction of WS, the best performing model was 2NN with the minimum MSE of 0.0310. This was approximately 0.0027 and 0.0093 lower than 3NN and 3NN with Increased nodes respectively.

b) For the prediction of VORP, all models give nearly identical results since the difference in error values is very less. Nevertheless, 3NN gave the best results with the least MSE of 0.0022 which is 0.0005 and 0.0001 lower than 2NN and 3NN with Increased nodes respectively.

c) Every model improved its prediction accuracy for the prediction of a player's VORP as compared to his WS. The average difference between the predictions is approximately 0.0283 for 2NN, 0.0315 for 3NN and 0.0380 for 3NN with Increased nodes. This proves that a 3 layered neural network with increased nodes improved the most for the prediction of VORP.

Finally, comparing our test results for MSE with the current state of the art that has an MSE of 4.32 (Nguyen Hoang Nguyen, 2021) for predicting WS, 2NN best improves the error by 4.2890. 3NN and 3NN with additional nodes improve it by 4.2863 and 4.2797 respectively.

Chapter 5 Conclusion

The main conclusion of our study along with improvements that can be made for further works in this domain is presented in this chapter. The first sections present the reflection analysis of our work done in the course of carrying this study and the main challenges encountered. The second section essentially describes the suggestions that might be added to upgrade the current work.

5.1 Reflection

Deep Learning presents many advantages when applied to the sports domain with an abundance of advanced and improved statistical data. However, due to the stochastic nature of Deep Learning models, even after doing repeated evaluation experiments the results fluctuated a little. This is expected as deep learning algorithms are still defined as a black box and presents a high amount of randomness than expected when compared to other algorithms. Nevertheless, despite the given lack of acute stability, the results indicate that applying Deep Learning to analyze an NBA player's performance has improved results. All the three models used to predict a player's performance by predicting his Win Shares improves the current state of the art in player performance prediction. Additionally, the performance judgement metric proposed in the study (VORP), when used to judge and predict a player's performance produces better results than the previously used metric. This signifies that using deep learning to predict a player's value to the team over a replacement player lowers the error in prediction and an organization can use this information to calculate how valuable a player is to a certain team, or what his trade value can be when compared to a replaced player.

5.2 Future Works

The improvements that can be made in the study to predict a player's performance include: -

- The features used in the study only cover the basic and advanced attributes; however, there are other improved versions of statistics that can be used to expand the dataset. This expanded dataset would present additional features for a Deep Learning model to learn on.
- The study only uses NBA seasons until 2017, the following year's information can be added to increase the number of instances in the dataset. This would increase the data provided to a DL model which would increase its prediction accuracy.
- Currently, the model training and validations use either the standard parameters or the parameters from the last study and for that reason, no hyperparameter tunning was applied. Tuning the hyperparameters of a model not only would benefit in optimizing its learning capability but also produces the optimal value of hyperparameters to a DL model which would result in more accurate predictions.
- Even though the evaluation experiments are repeated several times to reduce the randomness of DL models and produce the most stable results,

still the results may vary when the model is run again. Hence, the number of repetitions of evaluation experiments can be increased to present much more stable results.

- Basketball is a team sport, and many factors such as team cohesion, the coach of the team, shooting assistant, organization owner and its mental state may affect a player's performance. Using these features may produce more practical results.

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